MILITARY SPECIFICATION SHEET

WAVEGUIDE ASSEMBLIES, FLEXIBLE, NONTWISTABLE,
CLASS 2

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the waveguide assemblies described herein shall consist of this document and the latest issue of Specification MIL-W-287.

REQUIREMENTS: See table I.

Part number: See table I.

Extent of qualification: Qualification obtained for one assembly shall constitute qualification for other assemblies as specified in table II.

Cross reference information: See table III.

Substitution data: See table IV.

FSC 5985

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TABLE I. Flexible, nontwistable

Group number	1/ Part number M287/4-	TE_{10} mode frequency range (GHz)	Minimum centerline bending radius (inch)		Insertion loss (dB/ft)	Pressurization lb_f/in^2 gage	2/ VSWR (max) at			
			E plane	H plane			12°	24°	36°	>36°
I	01-XXXX 02-XXXX 03-XXXX 64-XXXX	1.12-1.70	13.02	27.07	0.01	15	1.05	1.06	1.07	1.10
	04-XXXX 05-XXXX 65-XXXX	1.70-2.60	9.51	14.00	0.015	20	1.06	1.07	1.08	1.10
	06-XXXX 07-XXXX 66-XXXX	2.20-3.30	5.00	9 75	0.018	25	1.06	1.07	1.08	1.10
	08-XXXX 09-XXXX 10-XXXX 11-XXXX 12-XXXX 13-XXXX 14-XXXX 15-XXXX 67-XXXX	2.60-3.95	4.50	6.38	0.02	30	1.08	1.09	1 10	1.12
II	16-XXXX 17-XXXX 18-XXXX 19-XXXX 20-XXXX 21-XXXX 22-XXXX 23-XXXX 68-XXXX	3.95-5.85	3.00	5.50	0 03	30	1.08	1.09	1.10	1.12
	24-XXXX 25-XXXX 26-XXXX 27-XXXX 28-XXXX 29-XXXX 30-XXXX 31-XXXX 32-XXXX 69-XXXX	5.85-8 20	2.50	5 00	0 05	30	1.08	1.09	1.10	1.12

See footnotes at end of table

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waveguide assemblies.

Flexure (degrees/ft) $1x10^5$ cycles		Jacket material	Retention force (lb_f)	Oper temp range -55 to	Mates with standard rigid waveguide MB5/1-	Flanges		
E plane	H plane					Flange 1 M3922/( )	Flange 2 M3922/( )	Material 3/
12.8	6.4	Neoprene	N/A	+100°C	017	58-007 (contact)	58-007 (contact)	CU
		Neoprene		+100°C	018	58-008 (contact)	58-008 (contact	AL
		Plastic		+65°C	017	58-007 (contact)	58-007 (contact)	CU
4/	4/	Neoprene		+100°C	017	58-007 (contact)	58-007 (contact)	CU
19.0	9.5	Neoprene	N/A	+100°C	031	58-009 (contact)	58-009 (contact)	CU
19.0	9.5				029	58-010 (contact)	58-010 (contact)	AL
4/	4/				031	58-009 (contact)	58-009 (contact)	CU
22.0	11.0	Neoprene	N/A	+100°C	037	58-011 (contact)	58-011 (contact)	CU
22.0	11.0				035	58-012 (contact)	58-012 (contact)	AL
4/	4/				037	58-011 (contact)	58-011 (contact)	CU
24.50	12.25	Neoprene	N/A	+100°C	043	64-001 (contact)	64-001 (contact)	CU
					041	64-002 (contact)	64-002 (contact)	AL
					043	56-001 (cover)	56-001 (cover)	CU
					041	56-002 (cover)	56-002 (cover)	AL
					043	61-002 (choke)	56-001 (cover)	CU
					041	61-001 (choke)	56-002 (cover)	AL
					043	61-002 (choke)	61-002 (choke)	CU
					041	61-001 (choke)	61-001 (choke)	AL
4/	4/				043	64-001 (contact)	64-001 (contact)	CU
43.0	21.5	Neoprene	N/A	+100°C	055	63-001 (contact)	63-001 (contact)	CU
					053	63-005 (contact)	63-005 (contact)	AL
					055	57-002 (cover)	57-002 (cover)	CU
					053	57-001 (cover)	57-001 (cover)	AL
					055	62-002 (choke)	57-002 (cover)	CU
					053	62-001 (choke)	57-001 (cover)	AL
					055	62-002 (choke)	62-002 (choke)	CU
					053	62-001 (choke)	62-001 (choke)	AL
4/	4/				055	63-001 (contact)	63-001 (contact)	CU
69.0	34.5	Neoprene	N/A	+100°C	067	63-002 (contact)	63-002 (contact)	CU
					065	63-006 (contact)	63-006 (contact)	AL
					067	55-001 (cover)	55-001 (cover)	CU
					065	55-002 (cover)	55-002 (cover)	AL
					067	60-001 (choke)	55-001 (cover)	CU
					065	60-002 (choke)	55-002 (cover)	AL
					067	60-001 (choke)	60-001 (choke)	CU
					065	60-002 (choke)	60-002 (choke)	AL
		Fluro-silicone rubber		+125°C	065	63-006 (contact)	63-006 (contact)	AL
4/	4/	Neoprene		+100°C	067	63-002 (contact)	63-002 (contact)	CU

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TABLE I. Flexible, nontwistable

Group number	1/ Part number M287/4-	TE_{10} mode frequency range (GHz)	Minimum centerline bending radius (inch)		Insertion loss (dB/ft)	Pressurization lb_f/in^2 gage	2/ VSWR (max) at			
			E plane	H plane			12°	24°	36°	>36°
II	33-XXXX 34-XXXX 35-XXXX 36-XXXX 37-XXXX 38-XXXX 39-XXXX 40-XXXX 70-XXXX	7.05-10.0	1.75	3.49	0.06	45	1.12	1.14	1.15	1.18
	41-XXXX 42-XXXX 71-XXXX	7.0-11.0	1.44	2.50	0.08	50	1.12	1.14	1.15	1.18
	43-XXXX 44-XXXX 45-XXXX 46-XXXX 47-XXXX 48-XXXX 49-XXXX 50-XXXX 72-XXXX	8 2-12.4	1.38	3.00	0.1	60	1.14	1.15	1.16	1.18
III	51-XXXX 52-XXXX 53-XXXX 54-XXXX 55-XXXX 56-XXXX 57-XXXX 73-XXXX	12.4-18.0	1.25	2.00	0.25	60	1.13	1.15	1.19	1.24
	58-XXXX 59-XXXX 60-XXXX 61-XXXX 62-XXXX 63-XXXX 74-XXXX	18.0-26.5	1 13	1 25	0.50	60	1.14	1.16	1.20	1.25

1/ The complete part number shall be the number shown and a four-digit number indicating the nominal relaxed length in inches.
2/ The VSWR value for any length between the lengths specified shall be the next higher value (e.g for a 15-inch length of M287/4-01, the VSWR would be 1.06·1).
3/ CU = copper alloy, AL = aluminum alloy.
4/ The assemblies are constructed from beryllium copper and the flexure cycle is 10^6 rather than 10^5. All other characteristics are as listed.

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waveguide assemblies - Continued.

Flexure (degrees/ft) $1x10^5$ cycles		Jacket material	Retention force (lbf)	Oper temp range -55 to	Mates with standard rigid waveguide MBS/1-	Flanges		
E plane	H plane					Flange 1 M3922/( )	Flange 2 M3922/( )	
98.50	49.25	Neoprene	N/A	+100°C	073	63-003 (contact)	63-003 (contact)	
					071	63-007 (contact)	63-007 (contact)	
					073	53-002 (cover)	53-002 (cover)	
					071	53-004 (cover)	53-004 (cover)	
					073	59-007 (choke)	53-002 (cover)	
					071	59-009 (choke)	53-004 (cover)	
					073	59-007 (choke)	59-007 (choke)	
4/	4/				071	59-009 (choke)	59-009 (choke)	
					073	63-003 (contact)	63-003 (contact)	
106.50	53.25	Neoprene	N/A	+100°C	155	70-001 (cover)	70-001 (cover)	
106.50	53.25				157	70-002 (cover)	70-002 (cover)	
4/	4/				155	70-001 (cover)	70-001 (cover)	
114.50	57.25	Neoprene	N/A	+100°C	079	63-004 (contact)	63-004 (contact)	
					077	63-008 (contact)	63-008 (contact)	
					079	53-001 (cover)	53-001 (cover)	
					077	53-003 (cover)	53-003 (cover)	
					079	59-006 (choke)	53-001 (cover)	
					077	59-008 (choke)	53-003 (cover)	
			100		079	59-006 (choke)	59-006 (choke)	
4/	4/		100		077	59-008 (choke)	59-008 (choke)	
			N/A		079	63-004 (contact)	63-004 (contact)	
114.50	57.25	Neoprene	N/A	+100°C	089	53-005 (cover)	53-005 (cover)	
					090	53-006 (cover)	53-006 (cover)	
					089	59-001 (choke)	53-005 (cover)	
					090	59-002 (choke)	53-006 (cover)	
					089	59-001 (choke)	59-001 (choke)	
		Plastic		+65°C	090	59-002 (choke)	59-002 (choke)	
4/	4/	Neoprene		+100°C	090	53-006 (cover)	53-006 (cover)	
					089	53-005 (cover)	53-005 (cover)	
137.50	68.75	Neoprene	N/A	+100°C	102	54-001 (cover)	54-001 (cover)	
					103	54-002 (cover)	54-002 (cover)	
					102	59-003 (choke)	54-001 (cover)	
					103	59-004 (choke)	54-002 (cover)	
					102	59-003 (choke)	59-003 (choke)	
					103	59-004 (choke)	59-004 (choke)	
4/	4/				102	54-001 (cover)	54-001 (cover)	

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TABLE II. Grouping for qualification.

Group number	Qualifying dash number	Dash numbers of assemblies qualified
I	05	01 thru 04, and 06 thru 15
I	65	64, 66 and 67
II	25	16 thru 24 and 26 thru 50
II	69	68, 70, 71 and 72
III	59	51 thru 58, and 60 thru 63
III	74	73

TABLE III. Cross reference (type designation to part number).

Type designation	Part number M287/4- ( )	Type designation	Part number M287/4- ( )
FG650BEEB	01	FG042BCCB	60
FG430BEEB	04	FG284BCCA	13
FG340BEEB	06	FG187BCCA	21
FG284BEEB	08	FG137BCCA	29
FG187BEEB	16	FG112BCCA	38
FG137BEEB	24	FG090BCCA	48
FG112BEEB	33	FG062BCCA	54
FG090BEEB	43	FG042BCCA	61
FG650BEEA	02	FG284BDDB	10
FG430BEEA	05	FG187BDDB	18
FG340BEEA	07	FG137BDDB	26
FG284BEEA	09	FG112BDDB	35
FG187BEEA	17	FG090BDDB	45
FG137BEEA	25	FG062BDDB	51
FG112BEEA	34	FG042BDDB	58
FG090BEEA	44	FG284BDDA	11
FG284BCCB	12	FG187BDDA	19
FG187BCCB	20	FG137BDDA	27
FG137BCCB	28	FG112BDDA	36
FG112BCCB	37	FG090BDDA	46
FG090BCCB	47	FG062BDDA	52
FG062BCCB	53	FG042BDDA	59

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TABLE IV. Substitution data. 1/

Dash Number	Manufacturer's code (FSCM)	Manufacturer's part number
01	80006	AF-2B-650
02	80006	AF-6B-650
04	80006	AF-2B-430
05	80006	AF-6B-430
06	80006	AF-2B-340
07	80006	AF-6B-340
08	80006	AF-12B-284
09	80006	AF-16B-284
10	80006	AS-2C-284
10	80006	AF-2B-284
11	80006	AS-6C-284
11	80006	AF-6B-284
12	80006	AS-3C-284
12	80006	AF-3B-284
13	80006	AS-7C-284
13	80006	AF-7B-284
14	80006	AS-4C-284
14	80006	AF-4B-284
15	80006	AS-8C-284
15	80006	AF-8B-284
16	80006	AF-12B-187
16	87864	TS-78268
17	80006	AF-16B-187
17	87864	TS-78269
18	80006	AF-2B-187
18	87864	TS-78271
19	80006	AF-6B-187
19	87864	TS-78272
20	80006	AF-3B-187
20	87864	TS-78230
21	80006	AF-7B-187
21	87864	TS-78270
22	80006	AF-4B-187
23	80006	AF-8B-187
24	80006	AF-12B-137
24	87864	TS-72377
25	80006	AF-16B-137
25	87864	TS-72378
26	80006	AS-2C-137
26	80006	AF-2B-137
26	87864	TS-72380
27	80006	AS-6C-137
27	80006	AF-6B-137
27	87864	TS-72381
28	80006	AS-3C-137
28	80006	AF-3B-137
28	87864	TS-72379
29	80006	AS-7C-137
29	80006	AF-7B-137
29	87864	TS-72145
30	80006	AS-4C-137
30	80006	AF-4B-137

See footnote at end of table.

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TABLE IV. Substitution data 1/ - Continued.

Dash Number	Manufacturer's code (FSCM)	Manufacturer's part number
31	80006	AS-6C-137
31	80006	AF-6B-137
33	80006	AF-12B-112
33	87864	TS-68828
34	80006	AF-16B-112
34	87864	TS-68781
35	80006	AS-2C-112
35	80006	AF-2B-112
35	87864	TS-68678
36	80006	AS-6C-112
36	80006	AF-6B-112
36	87864	TS-68829
37	80006	AS-3C-112
37	80006	AF-3B-112
37	87864	TS-68672
38	80006	AS-7C-112
38	80006	AF-7B-112
38	87864	TS-68634
39	80006	AS-4C-112
39	80006	AF-4B-112
40	80006	AS-6C-112
40	80006	AF-6B-112
41	80006	AF-2C-102
41	80006	AF-2B-102
42	80006	AF-6C-102
42	80006	AF-6B-102
43	80006	AF-12B-090
43	87864	TS-641606
44	80006	AF-16B-090
44	87864	TS-641607
45	80006	AF-2B-090
45	80006	AS-2C-090
45	87864	TS-641469
46	80006	AS-6C-090
46	80006	AF-6B-090
46	87864	TS-641377
47	80006	AS-3C-090
47	80006	AF-3B-090
47	87864	TS-641389
48	80006	AS-7C-090
48	80006	AF-7B-090
48	87864	TS-641372
49	80006	AS-4C-090
50	80006	AS-8C-090
51	87864	TS-57487
52	87864	TS-57603
53	87864	TS-57604
54	87864	TS-57483
58	87864	TS-49097
59	87864	TS-49083
60	87864	TS-49084
61	87864	TS-49082

1/ Assemblies covered by this specification sheet are substitutable for the manufacturer's part numbers shown This information in no way implies that the manufacturer's part is substitutable for the military part

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Navy - OS
Air Force - 17, 99
DLA - ES

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Army - AR
Navy - MC, CG, AS
Air Force - 11, 19

Preparing activity:
Navy - EC

Agent:
DLA - ES

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